

Class 6: Homework

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Write a function from the supplied code

```
# Can you improve this analysis code?
library(bio3d)
s1 <- read.pdb("4AKE") # kinase with drug
```

Note: Accessing on-line PDB file

```
s2 <- read.pdb("1AKE") # kinase no drug
```

Note: Accessing on-line PDB file

PDB has ALT records, taking A only, rm.alt=TRUE

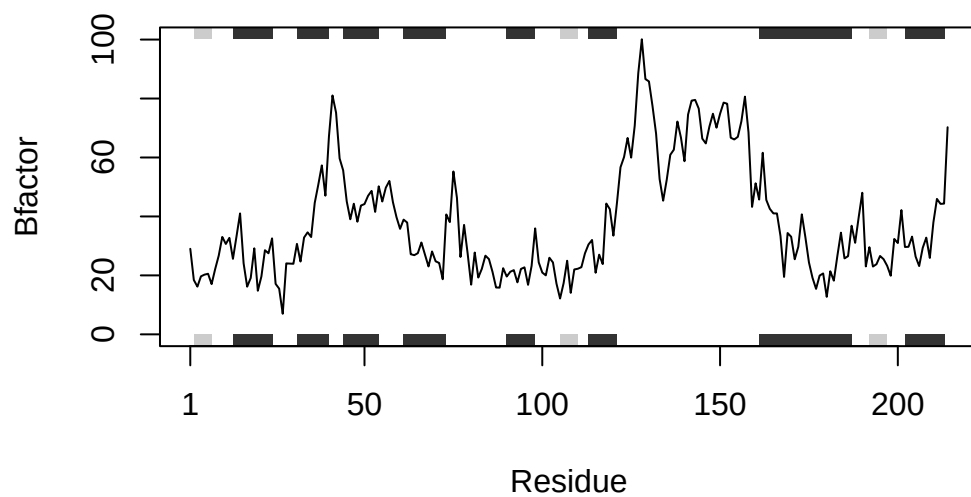
```
s3 <- read.pdb("1E4Y") # kinase with drug
```

Note: Accessing on-line PDB file

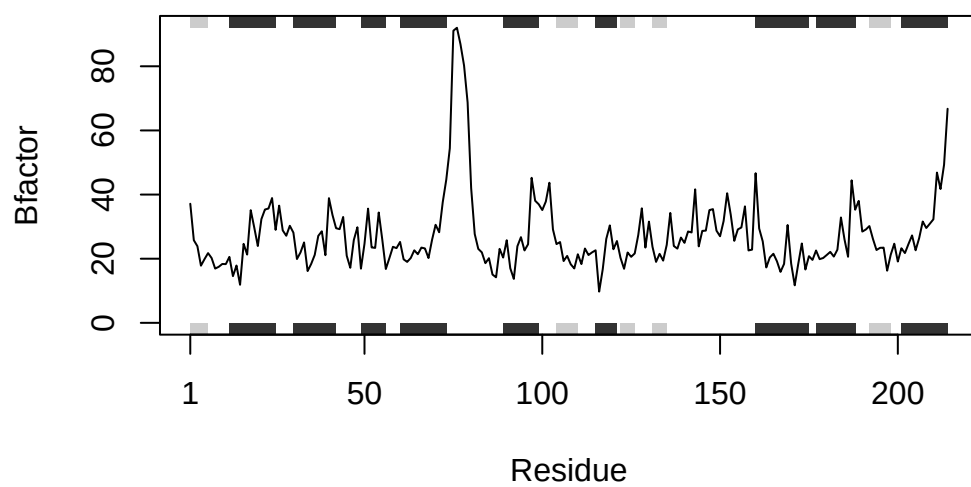
```
s1.chainA <- trim.pdb(s1, chain="A", elety="CA")
s2.chainA <- trim.pdb(s2, chain="A", elety="CA")
s3.chainA <- trim.pdb(s1, chain="A", elety="CA")

s1.b <- s1.chainA$atom$b
s2.b <- s2.chainA$atom$b
s3.b <- s3.chainA$atom$b

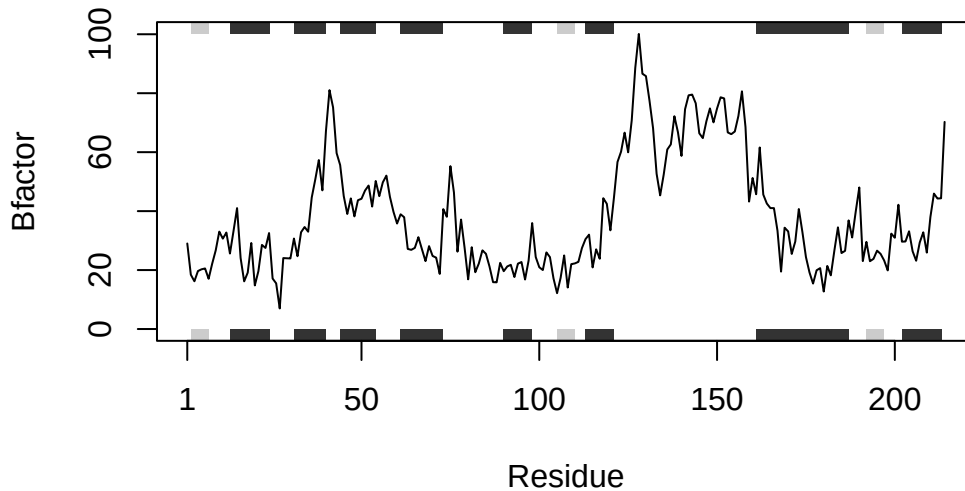
plotb3(s1.b, sse=s1.chainA, typ="l", ylab="Bfactor")
```



```
plotb3(s2.b, sse=s2.chainA, typ="l", ylab="Bfactor")
```



```
plotb3(s3.b, sse=s3.chainA, typ="l", ylab="Bfactor")
```



```
library(bio3d) #- Installed bio3d
read.protein.db <- function(pdb_ids) { # Here we assigned a function to a vector
  for (pdb_id in pdb_ids) { # Assigned our function to be plural

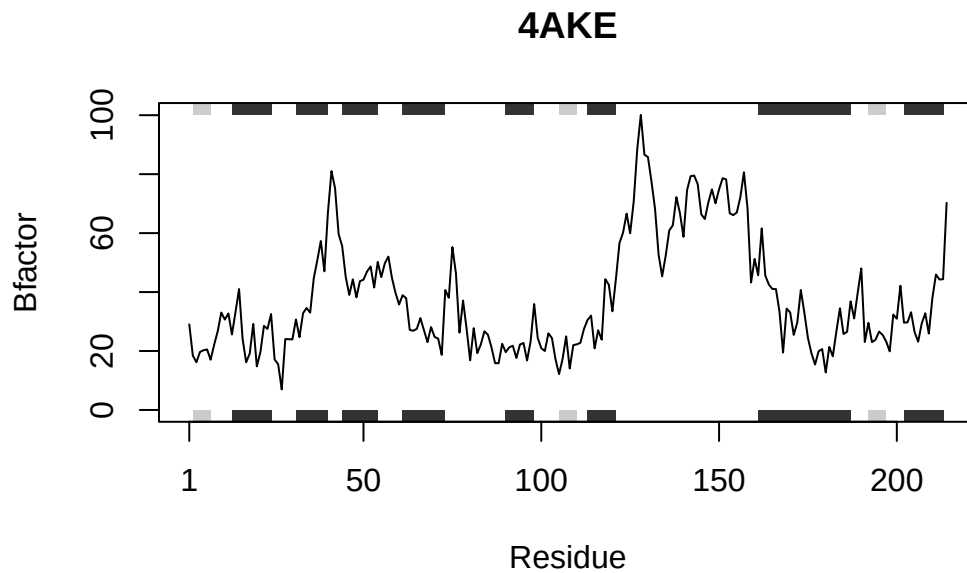
    pdb <- read.pdb(pdb_id) # - Assigning an object to equal what the function
#`read.pdb`does to our original argument
    chainA_pdb <- trim.pdb(pdb, chain="A", eley="CA") # Same thing here,
# assigning an object to a function that affects our original argument
    bfactor <- chainA_pdb$atom$b # - Same thing here, assigning an object
#to a function that affects our original argument

    plotb3(bfactor, sse=chainA_pdb, typ="l", ylab="Bfactor", main=pdb_id)
#Assigning an object to a specific column and row in our database,
#then plotting it, running main `main=pdb_id` to run all of our
#objects not just one.
  }
}
```

```
read.protein.db(c("4AKE", "1AKE", "1E4Y")) # - Printing our function.
```

Note: Accessing on-line PDB file

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
C:\Users\user\AppData\Local\Temp\Rtmps3E2PT/4AKE.pdb exists. Skipping download

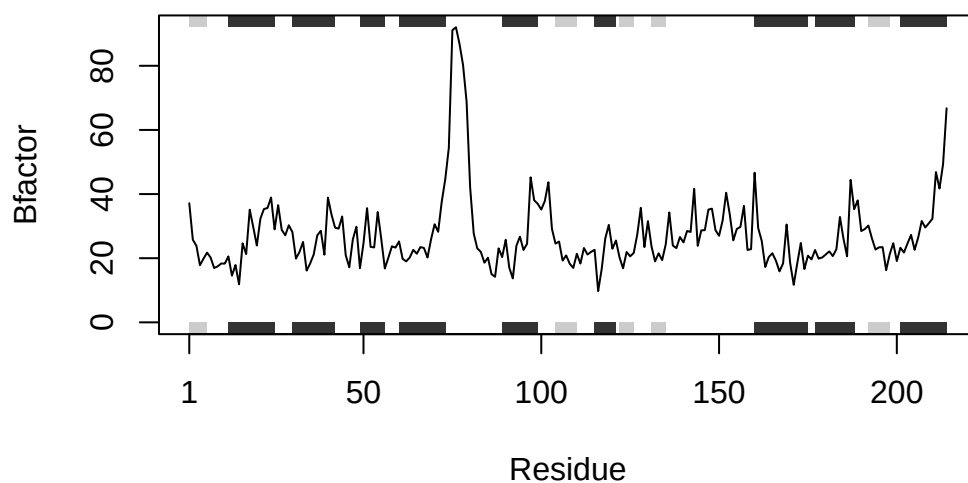


Note: Accessing on-line PDB file

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
C:\Users\user\AppData\Local\Temp\Rtmps3E2PT/1AKE.pdb exists. Skipping download

PDB has ALT records, taking A only, rm.alt=TRUE

1AKE



Note: Accessing on-line PDB file

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
C:\Users\user\AppData\Local\Temp\Rtmps3E2PT\1E4Y.pdb exists. Skipping download

1E4Y

